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VLSI Design Course Project
Designing and Implementing a 4-bit CLA and a SIMD-Reconfigurable 8-bit Adder

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Draft 1: Parts I and II

***Abstract*—This project designs a 4-bit carry look-ahead adder (CLA) in TSMC 180 nm, then joins two copies into a SIMD-reconfigurable 8-bit adder. The 4-bit adder uses a static CMOS prefix carry tree with OR-type transmit and dual-rail AOI22 sums, registered by TSPC flip-flops. Every gate, block and full circuit is simulated in NGSPICE before and after layout in MAGIC. The worst case is found by searching all single-input transitions rather than assumed: post-layout, the 4-bit adder's worst delay is 579.88 ps and the registered circuit works up to 1.147 GHz. In the 8-bit adder, a NAND2-and-inverter carry-gating cell switches between one 8-bit and two 4-bit additions, at a post-layout cost of 7.56% in carry delay and 1.50% in power.**

I. Introduction

Addition is the core of every datapath. A ripple-carry adder passes the carry from bit to bit, so the top sum bit waits for every lower bit and the delay grows with word length. A carry look-ahead adder (CLA) computes every carry from generate and propagate signals in parallel, so no carry waits for the one below it.

This project designs a registered 4-bit CLA in TSMC 180 nm (Part I), then joins two copies into an 8-bit adder that can be reconfigured into two independent 4-bit adders with one control signal (Part II). The primary objective throughout is the smallest *post-layout worst-case delay*; area and power are reported as trade-offs. Inputs are captured by D flip-flops on a rising clock edge and the result is captured on the next rising edge. Every output drives a $W_p/W_n = 20\lambda/10\lambda$ inverter, $\lambda = 0.09\,\mu$m, $V_{DD} = 1.8$ V.

All simulations use NGSPICE with the provided model; all layouts use MAGIC with the provided SCN6M_DEEP technology file. Every plot carries the title format `Name-Roll-question-part`, as the course asks.

II. Design Proposal

- $V_{DD} = 1.8$ V, technology TSMC 180 nm, $L_n = L_p = 2\lambda$.
- Logic style: static CMOS for every adder gate; TSPC dynamic flip-flops.

A. Generate, Propagate, Carry and Sum

For bits a_i and b_i:

$$g_i = a_i b_i, \qquad p_i = a_i \oplus b_i,$$

$$c_{i+1} = g_i + p_i c_i, \qquad s_i = p_i \oplus c_i.$$

Expanding the carry rule removes the ripple:

$$c_1 = g_0 + p_0 c_0$$
$$c_2 = g_1 + p_1 g_0 + p_1 p_0 c_0$$
$$c_3 = g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_0$$
$$c_4 = g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0 + p_3 p_2 p_1 p_0 c_0$$

Built directly, c_4 needs a five-input AND term, so a five-transistor stack, and c_0 fans out to every carry gate. Both make the circuit slow.

B. Optimization

OR-type transmit. In the carry terms, p_i can be replaced by $t_i = a_i + b_i$. The two differ only when $a_i = b_i = 1$, and then $g_i = 1$ already sets the carry. A NOR2 gives $\overline{t_i}$ in one light stage, and XOR is kept only for the sum. The idea follows Miao and Li [1].

Prefix carry tree. Bits are merged two at a time into group signals, and c_0 is added last:

$$G_{1:0} = g_1 + t_1 g_0, \quad T_{1:0} = t_1 t_0,$$
$$G_{3:0} = G_{3:2} + T_{3:2}\, G_{1:0}, \quad G_{2:0} = g_2 + t_2\, G_{1:0},$$
$$c_k = G_{k-1:0} + T_{k-1:0}\, c_0.$$

No gate has more than two transistors in series, and c_0 reaches each carry through one inverter and one gate.

TABLE I
Unit-size comparison of six candidate architectures, schematic, worst over all 4608 single-input transitions

Candidate	T	Worst (ps)	Hint vector (ps)
C1	208	771.3	421.4
C2	170	764.0	577.2
R	172	727.0	574.2
C3	170	796.4	623.5
C4	194	658.9	360.0
E	144	799.8	629.1

C1 is the textbook XOR and NAND-NAND design, C2 one complex gate per carry, R C2 with a mirror gate for c_4, C3 a design with stacks limited to 3, C4 the prefix tree, and E a

static Manchester chain. C4 was fastest, and remained fastest after sizing and layout, so it was chosen. The six designs differ only in their carry logic.

III. Design Details

Blocks. Each bit has a P/G front end: NAND2 gives $\overline{g_i}$, NOR2 gives $\overline{t_i}$, and two compound gates reuse them to make p_i (AOI21) and $\overline{p_i}$ (OAI21). The carry block is the prefix tree: OAI21 and NOR2 for pairs, AOI21 and NAND2 for groups down to bit 0, then one OAI21 per carry with $\overline{c_0}$. The sum is a dual-rail AOI22 XOR, $s_i = \overline{p_i c_i + \overline{p_i}\,\overline{c_i}}$, with an inverter for $\overline{c_i}$.

Sizing. The reference inverter is $W_n = 10\lambda$, $W_p = 20\lambda$. A unit-size gate matches its worst-case pull-up and pull-down resistance to that inverter. The final sizes were found by an automated search that minimised the worst delay over all transitions, then repeated with wire capacitance taken from the extracted layout. Table II lists the unit cells used in Sections IV and VI; Table III lists every instance of the final adder.

TABLE II
Unit cell sizes (widths in λ; several values are different stack positions)

Cell	T	NMOS W	PMOS W
Inverter	2	10	20
NAND2	4	20	20
NOR2	4	10	40
AOI21	6	10/20	40
OAI21	6	20	20/40
AOI22 XOR (sum)	10	10/20	20/40

TABLE III
Final adder sizing, every instance (widths in λ)

Inst.	Cell	Output	NMOS W	PMOS W
nand0	nand2	g0b	25	25
nor0	nor2	t0b	16	62
aoi0	aoi21	p0	4/8	16
oai0	oai21	p0b	8	8/16
nand1	nand2	g1b	25	25
nor1	nor2	t1b	20	78
aoi1	aoi21	p1	4/8	16
oai1	oai21	p1b	8	8/16
nand2	nand2	g2b	25	25
nor2	nor2	t2b	12	50
aoi2	aoi21	p2	8/16	32
oai2	oai21	p2b	16	16/32
nand3	nand2	g3b	20	20
nor3	nor2	t3b	12	50
aoi3	aoi21	p3	10/20	40
oai3	oai21	p3b	20	20/40
invc0	inv	c0b	10	20
g10	oai21_fn	g1_0	25	25/50
t10	nor2	t1_0	16	62
g32	oai21	g3_2	16	16/32
t32	nor2	t3_2	8	32
ig2	inv	g2	12	25
it2	inv	t2	20	39
g30	aoi21_fnp	g3_0b	4/8	16
t30	nand2	t3_0b	8	8
g20	aoi21_fnp	g2_0b	10/20	40
t20	nand2	t2_0b	25	25
ig10	inv	g1_0b	10	20
it10	inv	t1_0b	16	31
c1	oai21	c1	10	10/20
c2	oai21_fn	c2	25	25/50
c3	oai21_fn	c3	25	25/50
c4g	oai21_fnp	c4	16	16/32
x0	xor_aoi22	s0	5/25	10/50
x1	xor_aoi22	s1	6/20	13/40
x2	xor_aoi22	s2	12/16	25/32
x3	xor_aoi22	s3	10	20

D flip-flop. A true single-phase clock (TSPC) flip-flop [2] needs one clock and 11 transistors. At a 15 fF load it gave clock-to-Q plus setup of 212.9 ps, against 316.8 ps for a 22-transistor transmission-gate master-slave flip-flop, so TSPC was chosen.

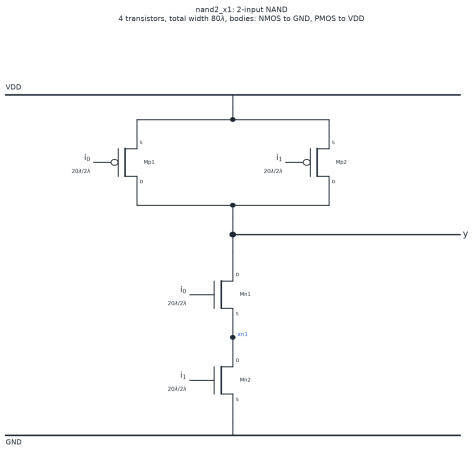


Fig. 3. Transistor schematic: NAND2.

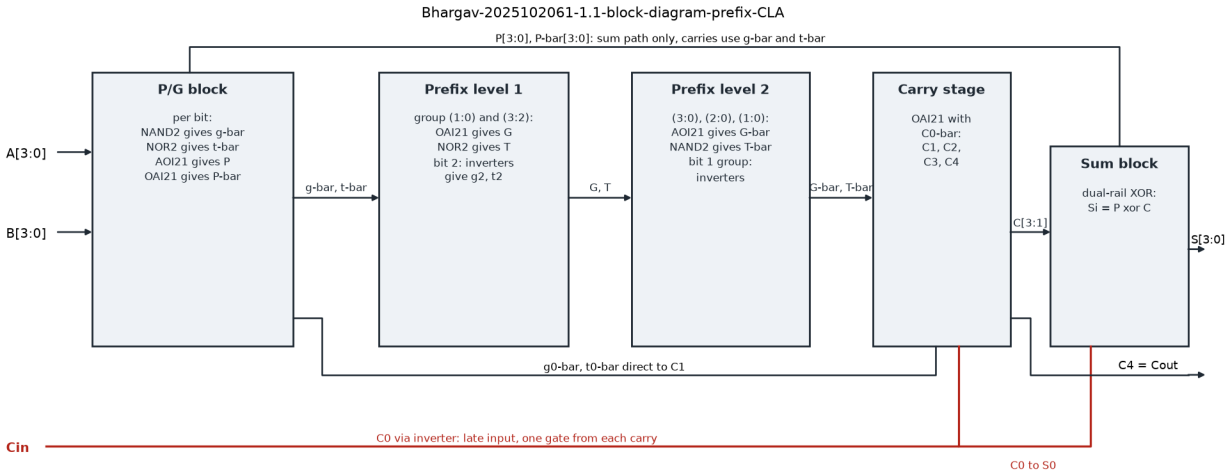


Fig. 1. Block-level architecture of the 4-bit CLA.

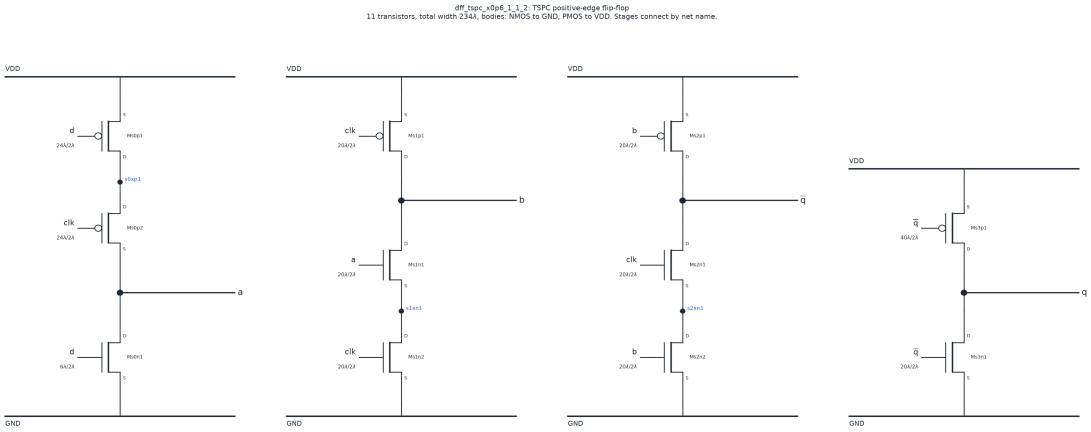


Fig. 2. TSPC flip-flop, stage drives x0.6, x1, x1, x2, sized at the largest adder input load of 47.76 fF.

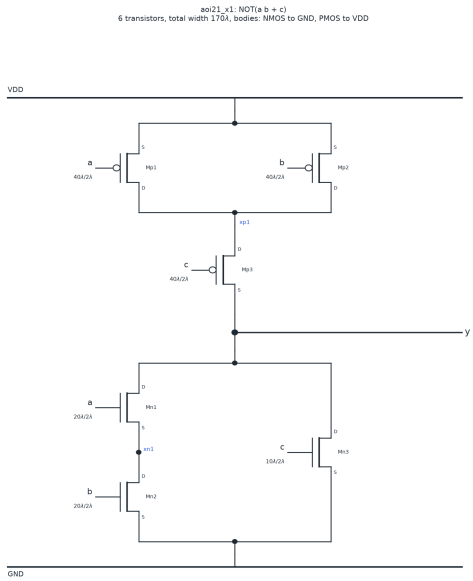


Fig. 4. Transistor schematic: AOI21, used for p_i and the group generates.

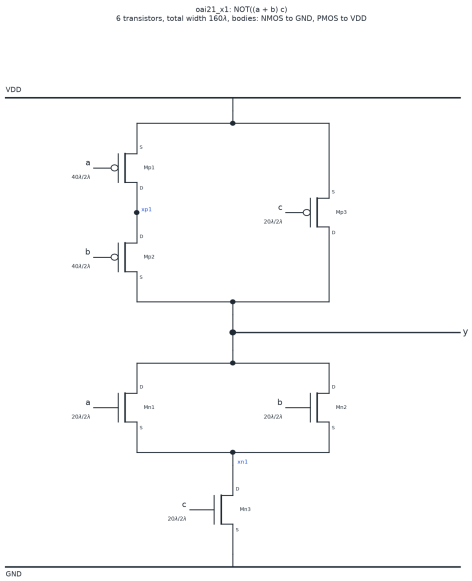


Fig. 5. Transistor schematic: OAI21, used for $\overline{p_i}$ and the final carries.

IV. NGSPICE Simulation (Pre-Layout)

Each gate is driven through two inverters, so its inputs have realistic edges, and its output drives the $20\lambda/10\lambda$ inverter. Inputs follow an Eulerian walk, which applies every single-input change from every state. Delay is measured from the 50% point of the changing input to the last 50% crossing of the output.

A. Inverter

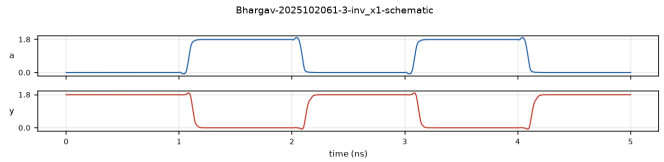


Fig. 6. Inverter plot: pre-layout.

B. NAND2

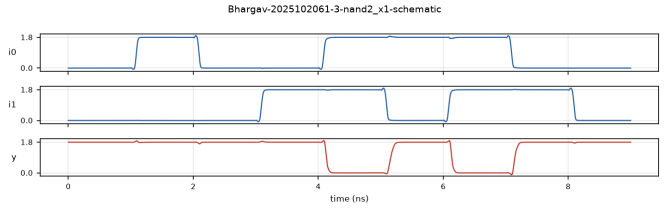


Fig. 7. NAND2 plot: pre-layout.

C. NOR2

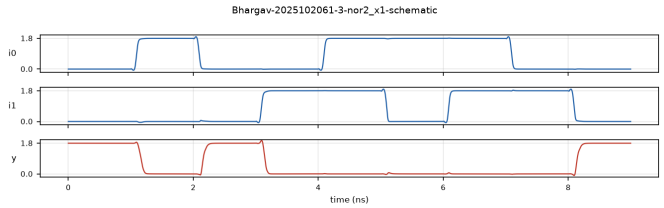


Fig. 8. NOR2 plot: pre-layout.

D. AOI21

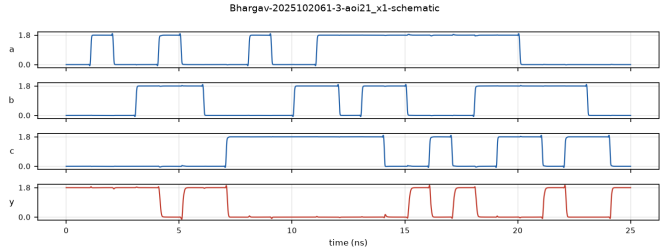


Fig. 9. AOI21 plot: pre-layout.

E. OAI21

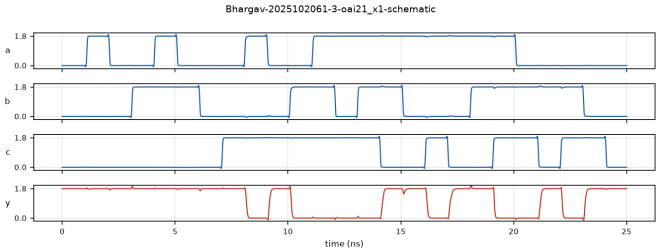


Fig. 10. OAI21 plot: pre-layout.

F. AOI22 XOR (sum)

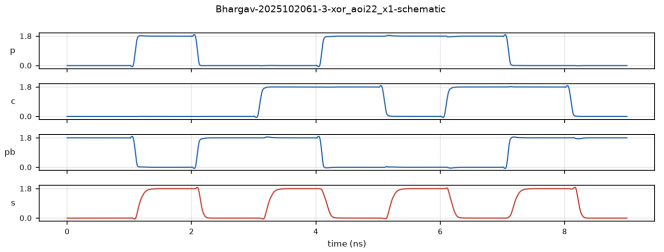


Fig. 11. AOI22 XOR (sum) plot: pre-layout.

G. D Flip-Flop

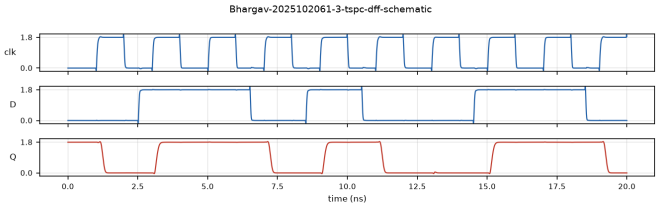


Fig. 12. TSPC D flip-flop plot: pre-layout. Q takes D at each rising clock edge.

H. P/G Block

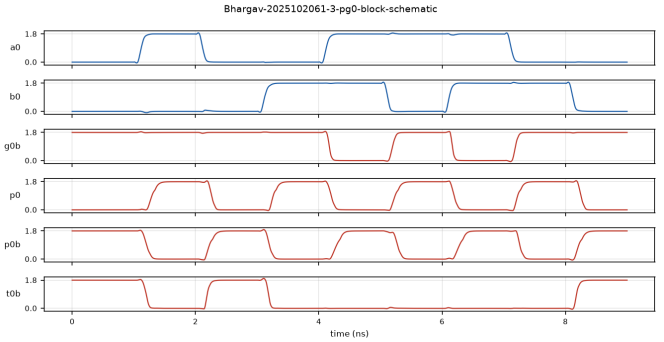


Fig. 13. P/G Block plot: pre-layout.

I. Carry Block

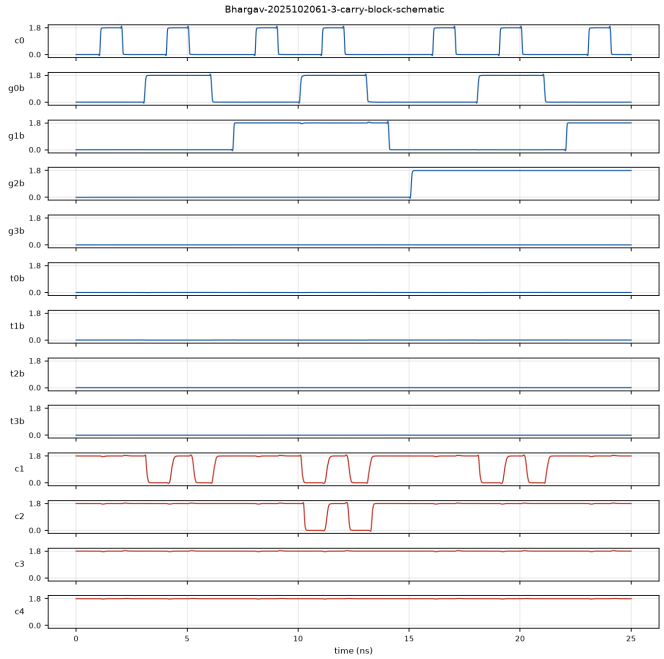


Fig. 14. Carry Block plot: pre-layout.

J. Sum Block

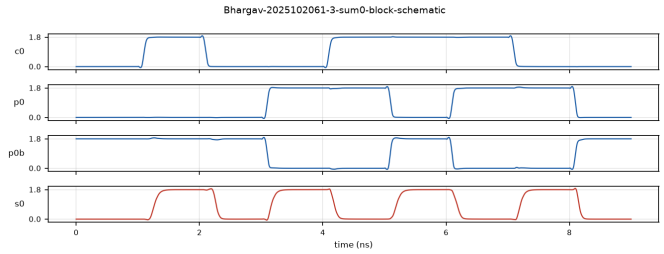


Fig. 15. Sum Block plot: pre-layout.

TABLE IV
GATE DELAYS, PRE-LAYOUT, WORST OVER ALL SINGLE-INPUT CHANGES

Gate	tpLH (ps)	tpHL (ps)	Errors
Inverter	38.59	35.7	0
NAND2	66.02	45.44	0
NOR2	64.35	70.61	0
AOI21	91.52	74.76	0
OAI21	92.91	75.84	0
AOI22 XOR (sum)	134.99	122.89	0

V. D FLIP-FLOP CHARACTERIZATION FROM NGSPICE

A. TSPC Flip-Flop

The flip-flop is characterised at its real load, 47.76 fF, with both data values. Definitions follow Stojanović and Oklobdžija [3].

B. Clock to Q Delay t_{cq}

Time from the 50% point of the rising clock to the 50% point of Q, with data set up long before the edge.

TABLE V
CLOCK TO Q DELAY, PRE-LAYOUT

Parameter	Time (ps)
t_{cq}, Q rising	134.1
t_{cq}, Q falling	208.5
t_{cq} (worst)	208.5

C. Set Up Time t_{su}

The shortest D-to-clock time for which clock-to-Q stays within 10% of its long-setup value.

TABLE VI
SET UP TIME, PRE-LAYOUT

Parameter	Time (ps)
t_{su}, D rising	-26.0
t_{su}, D falling	34.0
t_{su} (worst)	34.0

D. Hold Time t_{hold}

The shortest clock-to-D time for which a later data change leaves Q correct, with no disturbance above 10% of V_{DD}.

TABLE VII
HOLD TIME, PRE-LAYOUT

Parameter	Time (ps)
t_{hold}, D rising	-11.7
t_{hold}, D falling	55.1
t_{hold} (worst)	55.1

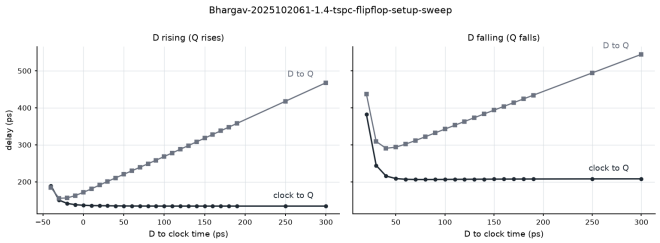


Fig. 16. Clock-to-Q and D-to-Q delay against D-to-clock time, schematic.

VI. MAGIC LAYOUTS

Every layout below is a screenshot of MAGIC's own layout window. Each cell passes MAGIC's DRC with zero errors, and each extracted netlist matches the schematic in LVS, using a comparison script written for this project because netgen was not available. Post-layout plots use the netlist MAGIC extracted, with all parasitic capacitance.

A. Inverter

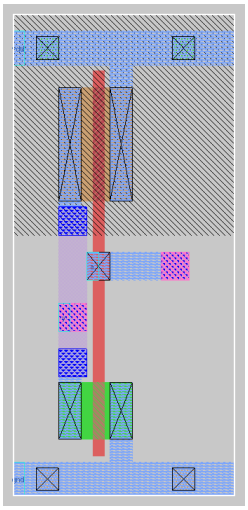


Fig. 17. MAGIC layout of Inverter.

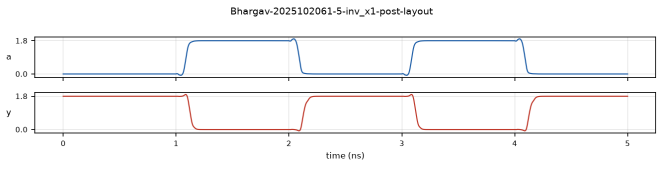


Fig. 18. Inverter plot: post-layout.

B. NAND2

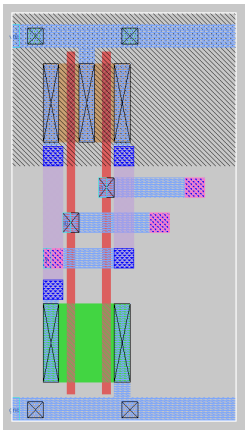


Fig. 19. MAGIC layout of NAND2.

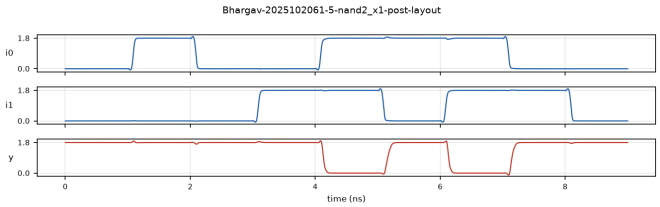


Fig. 20. NAND2 plot: post-layout.

C. NOR2

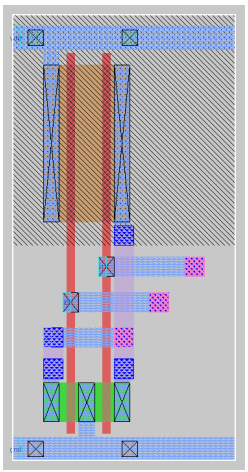


Fig. 21. MAGIC layout of NOR2.

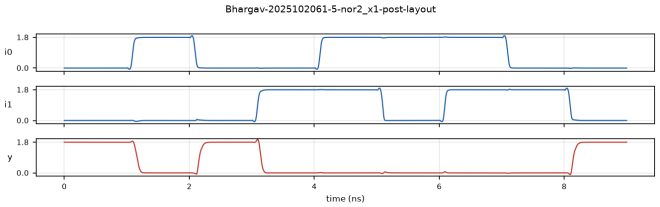


Fig. 22. NOR2 plot: post-layout.

D. AOI21

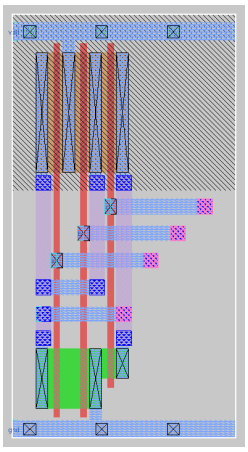


Fig. 23. MAGIC layout of AOI21.

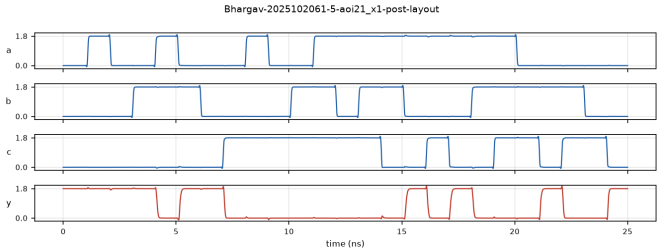


Fig. 24. AOI21 plot: post-layout.

E. OAI21

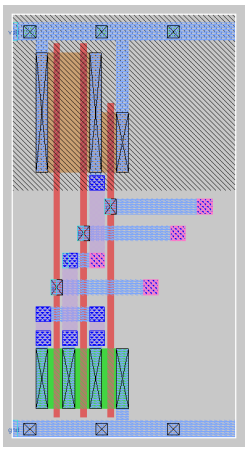


Fig. 25. MAGIC layout of OAI21.

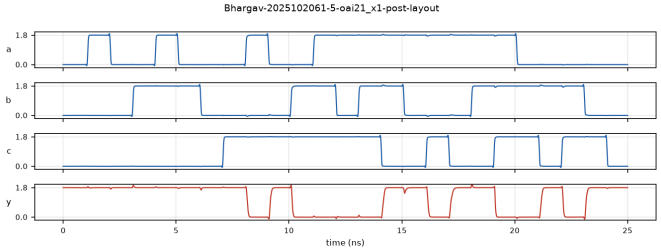


Fig. 26. OAI21 plot: post-layout.

F. AOI22 XOR (sum)

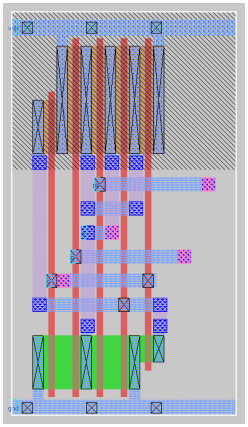


Fig. 27. MAGIC layout of AOI22 XOR (sum).

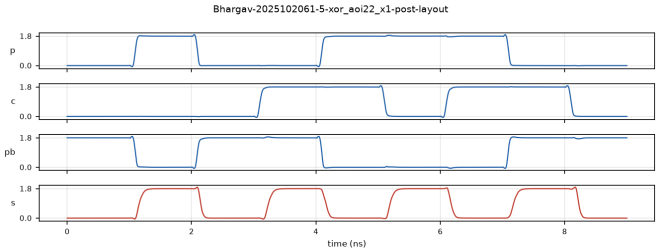


Fig. 28. AOI22 XOR (sum) plot: post-layout.

G. D Flip-Flop

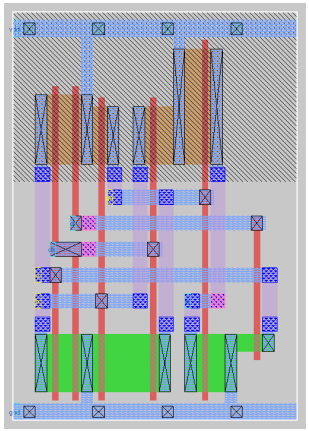


Fig. 29. MAGIC layout of the TSPC D flip-flop.

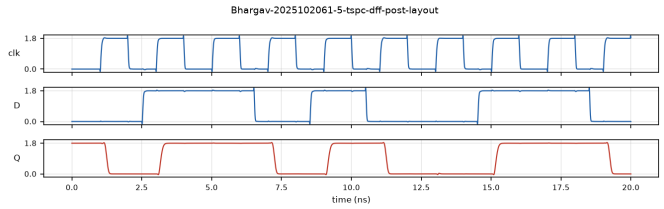


Fig. 30. TSPC D flip-flop plot: post-layout.

H. Clock to Q, Set Up and Hold Time of the Flip-Flop

TABLE VIII
FLIP-FLOP TIMING, PRE- AND POST-LAYOUT

Parameter (ps)	Pre-layout	Post-layout
t_{cq}	208.5	204.2
t_{su}	34.0	38.0
t_{hold}	55.1	45.7

I. P/G Block

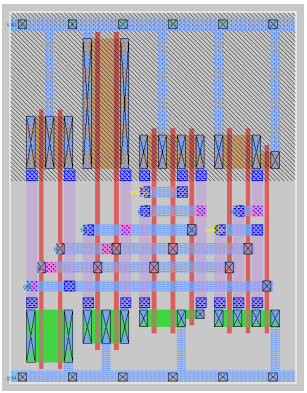


Fig. 31. MAGIC layout of the P/G Block.

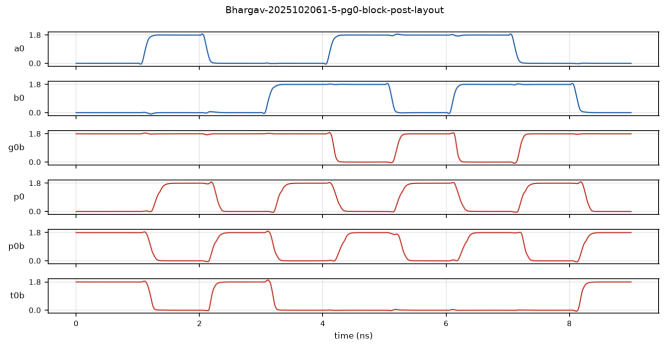


Fig. 32. P/G Block plot: post-layout.

J. Carry Block

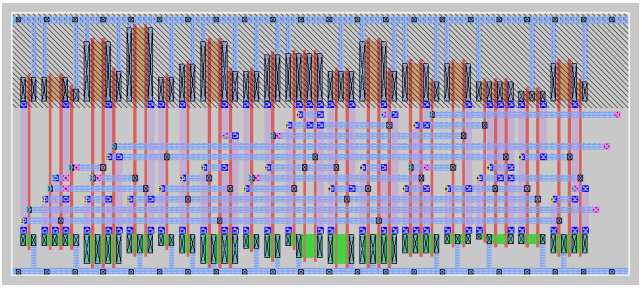


Fig. 33. MAGIC layout of the Carry Block.

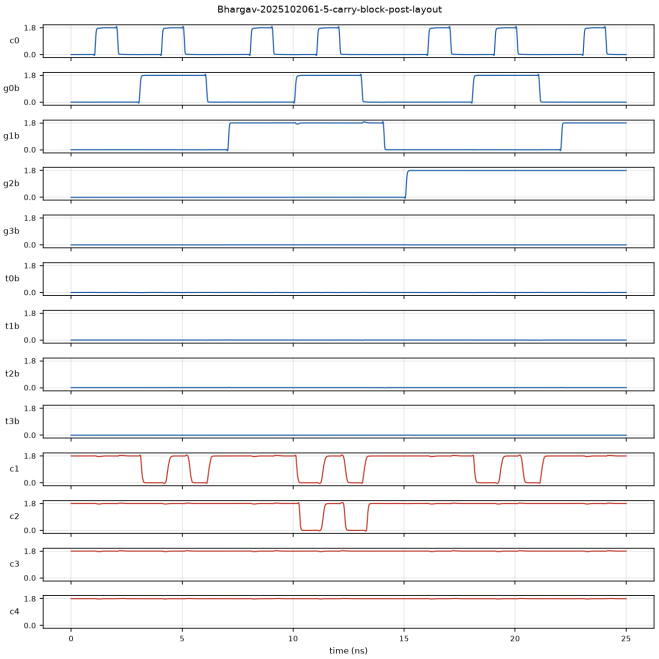


Fig. 34. Carry Block plot: post-layout.

K. Sum Block

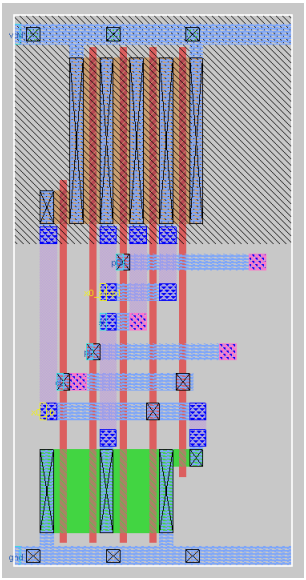


Fig. 35. MAGIC layout of the Sum Block.

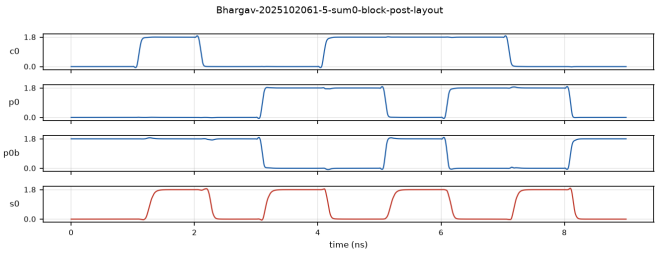


Fig. 36. Sum Block plot: post-layout.

TABLE IX
GATES AND BLOCKS: PRE-LAYOUT AGAINST POST-LAYOUT. BLOCK DELAYS ARE WORST OVER ALL THEIR SINGLE-INPUT TRANSITIONS.

Cell	Pre tpLH	Pre tpHL	Post tpLH	Post tpHL
Inverter	38.59	35.7	39.4	36.47
NAND2	66.02	45.44	66.14	45.9
NOR2	64.35	70.61	64.67	70.81
AOI21	91.52	74.76	88.88	71.15
OAI21	92.91	75.84	84.86	69.97
AOI22 XOR (sum)	134.99	122.89	135.36	122.15
block pg0	187.21	168.95	201.02	176.08
block pg1	187.4	171.1	200.97	169.7
block pg2	195.19	176.14	206.87	180.24
block pg3	199.86	189.09	210.24	190.19
block carry	319.87	306.42	335.79	336.37
block sum0	176.33	173.69	182.49	171.72
block sum1	157.14	144.1	161.33	142.78
block sum2	134.23	119.06	134.85	118.7
block sum3	134.71	114.42	137.22	117.54

Pre- and post-layout plots match for every gate and block. Delays change little at cell level, because a single cell has short wires; the difference grows once the cells are wired together (Section XI).

VII. NGSPICE CIRCUIT (PRE-LAYOUT)

The full circuit of the course's Figure 1(i) has 9 input flip-flops (a, b and c_0), the adder and 5 output flip-flops. A ninth flip-flop on c_0 keeps every input on the same clock. Inputs change before a rising edge; the sum appears at the output flip-flops one clock later.

Worst-case delay. The worst case was found by search, not assumed: all 4608 single-input transitions, each timed to the last 50% crossing so false sums cannot shorten it. Pre-layout worst: **516.30 ps**, a1 rise to s3 rise, A=0101 B=0001 C0=0. The course hint vector, $A = 1111$, $B = 0000$ with c_0 toggling, gives only 377.45 ps, because c_0 enters the prefix tree last.

Clock. The shortest clock at which the pre-layout registered circuit gave no wrong outputs on 264 vectors is 740.2 ps, 1.351 GHz.

VIII. MAGIC LAYOUT PLAN

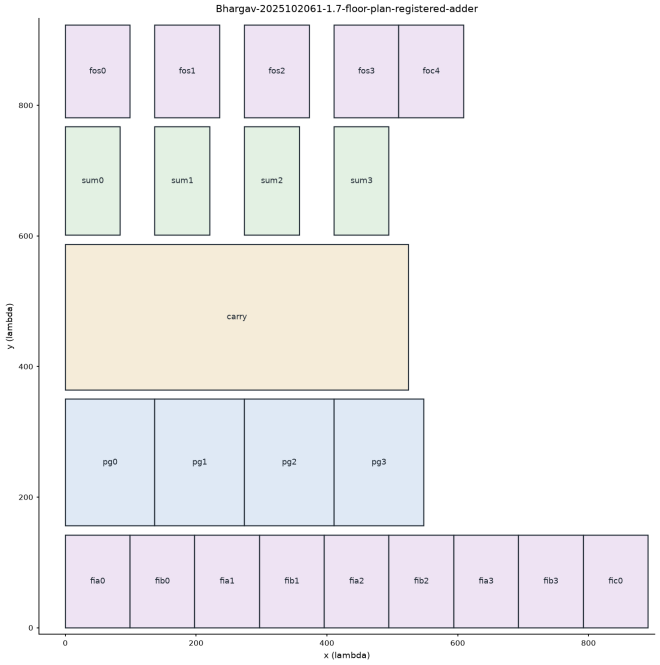


Fig. 38. MAGIC layout floor plan. Rows from the bottom: input flip-flops, P/G bits, carry block, sum bits, output flip-flops.

Signals flow upward through the rows. The P/G and sum cells sit on one bit pitch, so each sum bit lines up with its own P/G bit.

TABLE X
PITCHES

Pitch	Value (λ)	Value (μm)
Horizontal, bit pitch	137	12.33
Vertical, row pitches	156, 208, 237, 180	14.04, 18.72, 21.33, 16.20
Routing grid, metal 3 and 4	9	0.81

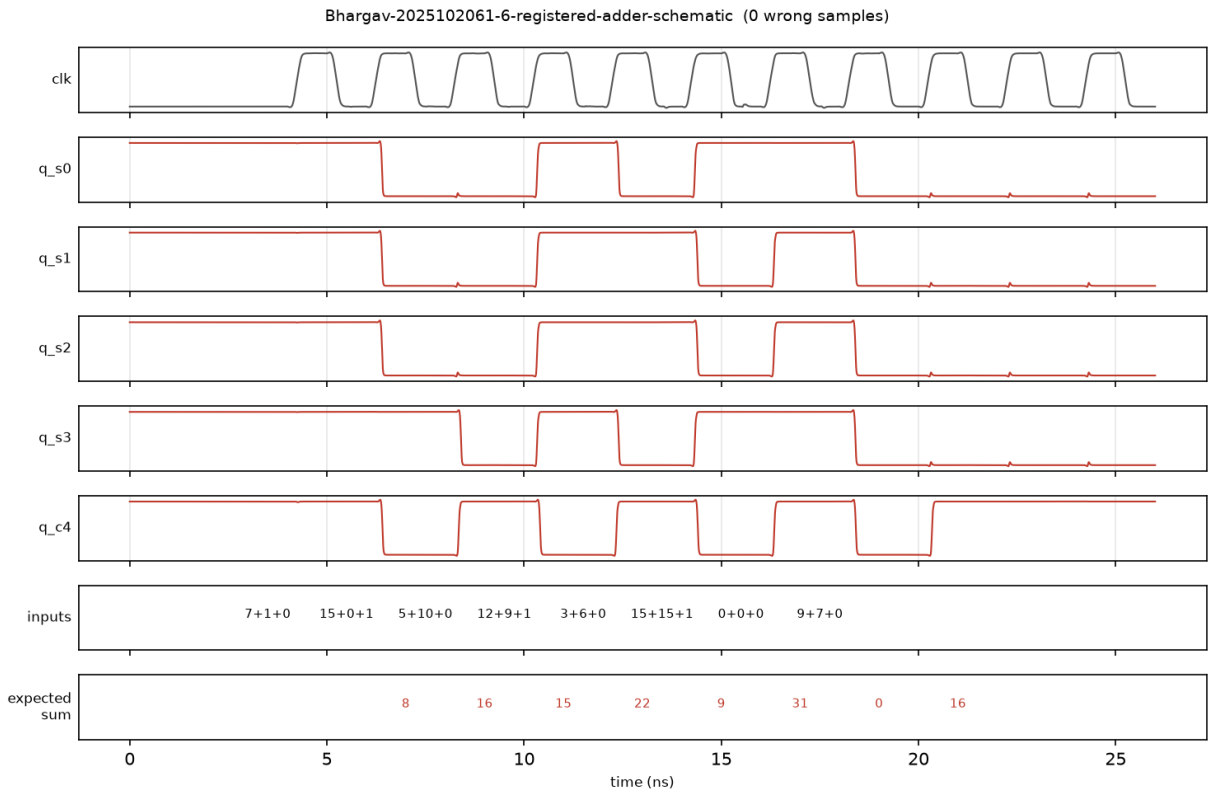


Fig. 37. Registered adder plot: pre-layout. Each sum appears one clock after its inputs.

IX. Final MAGIC Layout

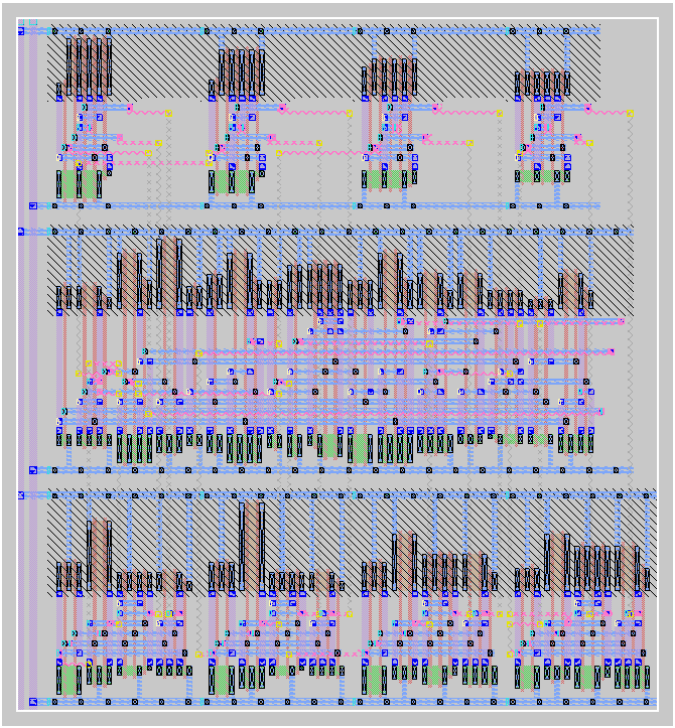


Fig. 39. Final 4-bit adder layout in MAGIC.

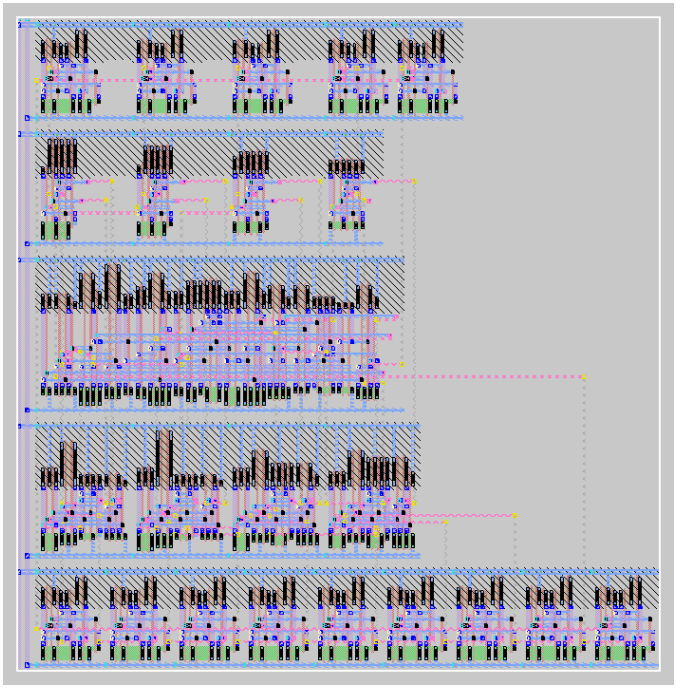


Fig. 40. Full registered circuit layout in MAGIC, with input and output flip-flops.

TABLE XI
LAYOUT RESULTS

Layout	DRC	LVS	Size (λ)	Area (μm^2)
Adder	0	match	582x623	2936.9
Registered	0	match	925x935	7005.5

The adder has 194 transistors and the registered circuit 348. Extracted wire capacitance is 376.9 fF for the adder and 624.5 fF for the registered circuit.

X. Post-Layout Simulations and Delays

A. Critical Path Delay

TABLE XII
COURSE HINT VECTOR, $A = 1111$, $B = 0000$, c_0 TOGGLING, POST-LAYOUT

c_0 edge	Output	Type	Delay (ps)
rise	S0	tpHL	72.8
rise	S1	tpHL	446.81
rise	S2	tpHL	402.78
rise	S3	tpHL	396.98
rise	C4	tpLH	300.03
fall	S0	tpLH	102.04
fall	S1	tpLH	454.15
fall	S2	tpLH	385.19
fall	S3	tpLH	385.81
fall	C4	tpHL	283.28

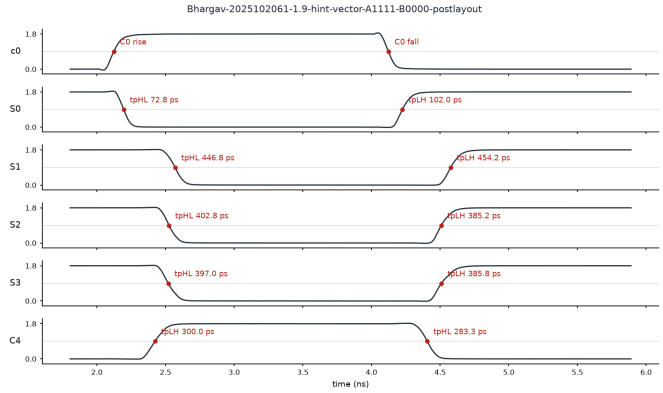


Fig. 42. Hint vector, extracted netlist, 50% crossings marked.

The slowest hint-vector output takes 454.1 ps, but the search over all transitions finds a slower one: **579.88 ps**, a1 rise to s3 rise, A=0001 B=0101 C0=0. When several inputs change together, a sample of 65520 transitions re-simulated on the extracted netlist gives **589.85 ps**. This is taken as $t_{pd,max}$.

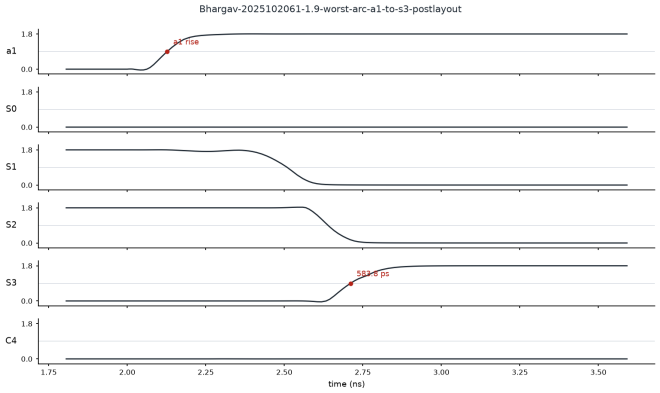


Fig. 43. Worst single-input transition, extracted netlist.

B. Maximum Clock Frequency

The course formula gives

$$T = t_{pd,max} + t_{su} + t_{hold} = 589.85 + 38.0 + 45.7 = 673.6\,\text{ps},$$

$f_{max} = 1.4847\,\text{GHz}$. At that clock the extracted registered circuit gave 264 wrong output samples, because the formula leaves out clock-to-Q. Using $T \geq t_{cq} + t_{pd,max} + t_{su} = 832.0\,\text{ps}$ instead, and checking by simulation, the shortest clock that works is **872.1 ps, 1.147 GHz**.

C. Power

$$P_{avg} = \frac{V_{DD}}{T_2 - T_1} \int_{T_1}^{T_2} I_{VDD}(t)\,dt$$

At the verified clock of 872.1 ps, over 264 cycles of random and critical vectors, the circuit's own supply gives $P_{avg} = 3.3899\,\text{mW}$, so $PDP = P_{avg} \times t_{pd,max} = 1999.51\,\text{fJ}$. Input drivers, clock buffers and load inverters use a separate supply.

TABLE XIII
COMPARISON OF PRE- AND POST-LAYOUT

Parameter	Pre-Layout	Post-Layout
Worst single-input delay	516.30 ps	579.88 ps
Hint vector delay	377.45 ps	455.23 ps
t_{cq}	208.5 ps	204.2 ps
t_{su}	34.0 ps	38.0 ps
t_{hold}	55.1 ps	45.7 ps
$T_{clk}(min)$, verified	740.2 ps	872.1 ps
$f_{clk}(max)$, verified	1.351 GHz	1.147 GHz
P_{avg} at the post-layout clock	3.0883 mW	3.3899 mW

Post-layout delays are longer, by 12.3% for the worst single-input delay, because the extracted netlist adds the capacitance of every wire and junction. Flip-flop clock-to-Q drops slightly after layout, because shared diffusion in the layout has less capacitance than the schematic's estimate.

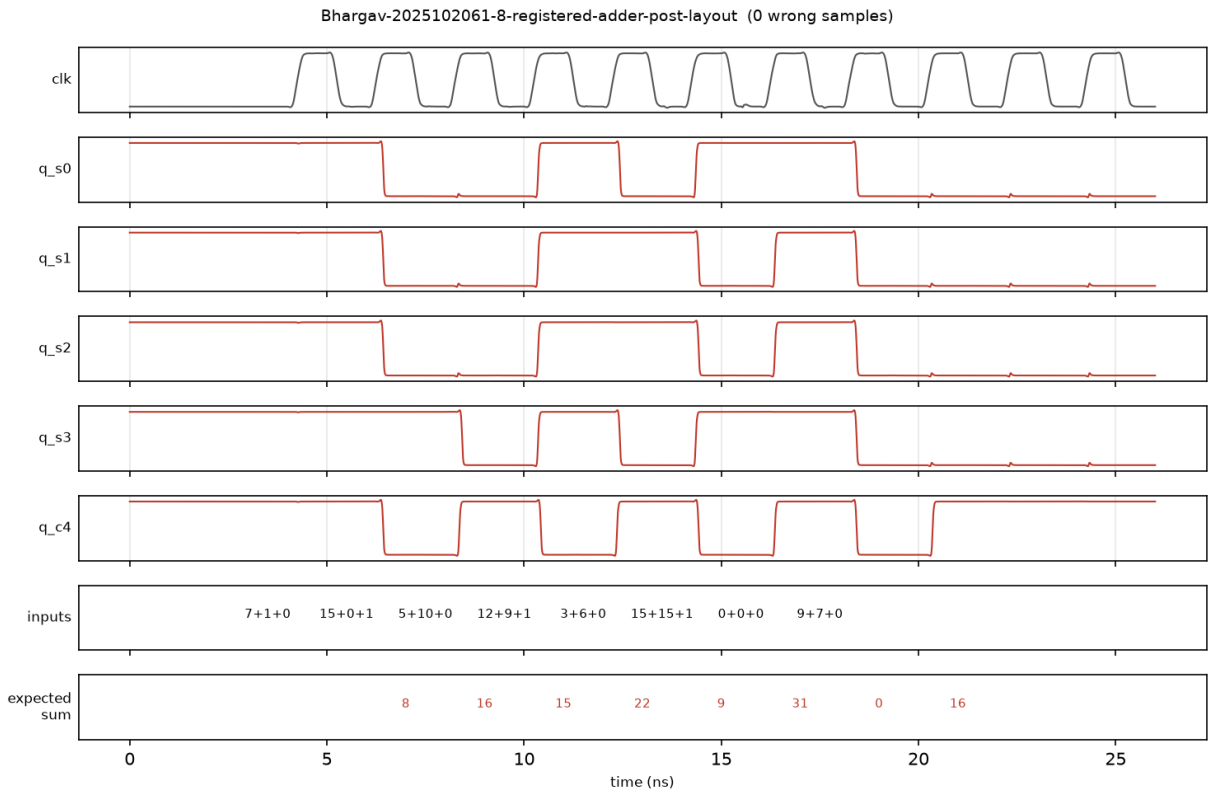


Fig. 41. Registered adder plot: post-layout. The outputs match the pre-layout run.

D. Area-Delay Trade-off

TABLE XIV
UNIFORM SCALING OF THE FINAL SIZES, POST-LAYOUT

Scale	Delay (ps)	Area (μm^2)	P (mW)
0.7	602.38	2574.0	0.9832
0.85	601.39	2743.7	1.1156
1.0	579.88	2936.9	1.2704
1.2	590.99	3293.8	1.4821

Bigger is not always faster: scaling up by 1.2 makes the extra load outweigh the extra drive. MAGIC's resistance extraction crashed on this layout, so resistance was bounded instead: with the longest poly run's resistance in series with every gate, the worst delay rises by at most 3.61%.

XI. Part II: SIMD-Reconfigurable 8-bit Adder

GPUs often split one wide datapath into narrower sub-words [5]. Here two copies of the Part I adder, CLA0 for bits 3..0 and CLA1 for bits 7..4, form an 8-bit adder with a control input SIMD_MODE:

- SIMD_MODE $= 0$: CLA0's carry-out goes into CLA1's carry-in, one 8-bit add.
- SIMD_MODE $= 1$: the inter-block carry is forced to 0, two independent 4-bit adds.

The cost of this is measured against a *baseline*: the same two adders wired directly, with no gating.

A. Architecture

The block diagram below shows the two Part I adders and the carry-gating cell between them. CLA0's carry-out, c4lo, is also an output, because in dual mode it is the carry of the lower 4-bit result; it drives the same $20\lambda/10\lambda$ load as the other outputs. The baseline is the same diagram without the gate, with c4lo wired straight to CLA1's carry-in.

B. Carry-Gating Cell

$$c_{in,hi} = c_{out,lo} \cdot \overline{\text{SIMD_MODE}}$$

This is an AND gate. In static CMOS the cell is a NAND2 followed by an inverter on the carry path, with a unit inverter making $\overline{\text{SIMD_MODE}}$; the mode is held during an add, so that inverter is off the timing path. The cell has 8 transistors.

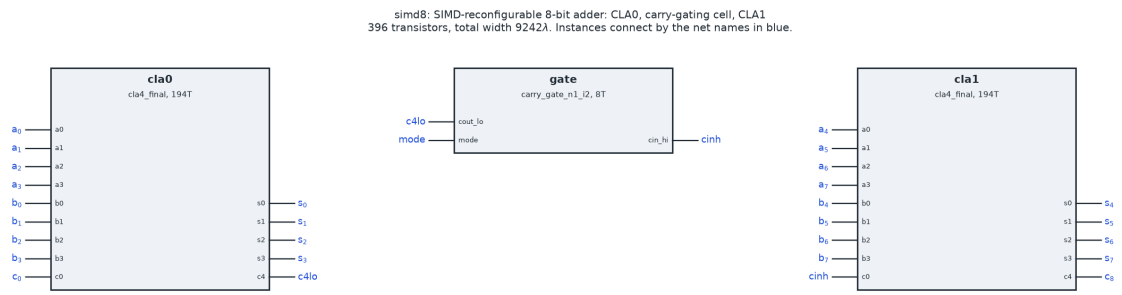


Fig. 44. SIMD 8-bit adder: CLA0, the carry-gating cell and CLA1. c4lo is CLA0's carry-out and cinh CLA1's carry-in.

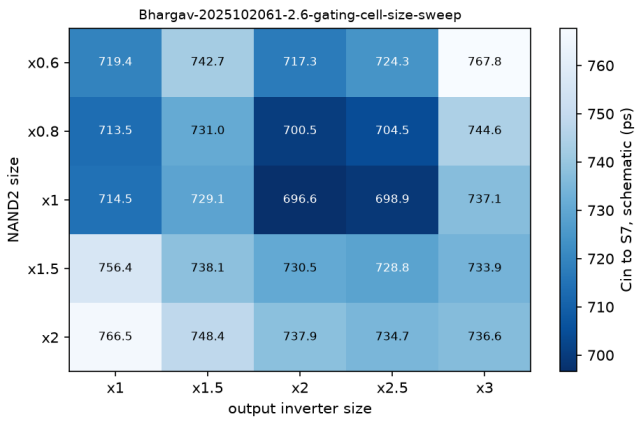


Fig. 45. Gate sizing sweep, Cin to S7 in mode 0, schematic. Best: NAND2 x1, inverter x2.

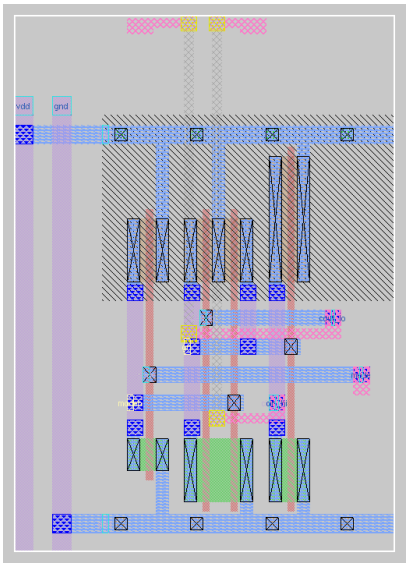


Fig. 46. MAGIC layout of the carry-gating cell, with carry pads above it.

TABLE XV
CARRY-GATING CELL ALONE, COUT_LO TO CIN_HI WITH THE 20/10λ LOAD

View	Worst	Rising	Falling
Pre-layout	104.27	89.12	104.27
Post-layout	110.46	95.2	110.46

C. Layout

The Part I adder was laid out again with two extra metal 3 pads above its top row, for c_4 and c_0. The 8-bit layout places this cell twice, with the gating cell between the copies in the SIMD design, and joins the inter-block carries on metal 5, which no cell uses. A first attempt that placed all gates of both adders in one set of rows ran out of routing tracks.

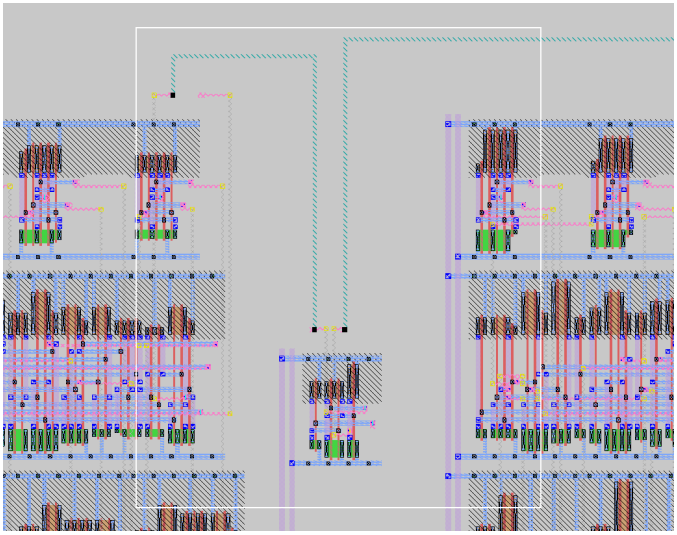


Fig. 48. Close-up in MAGIC: the gating cell between the two adders, with its two metal 5 carries.

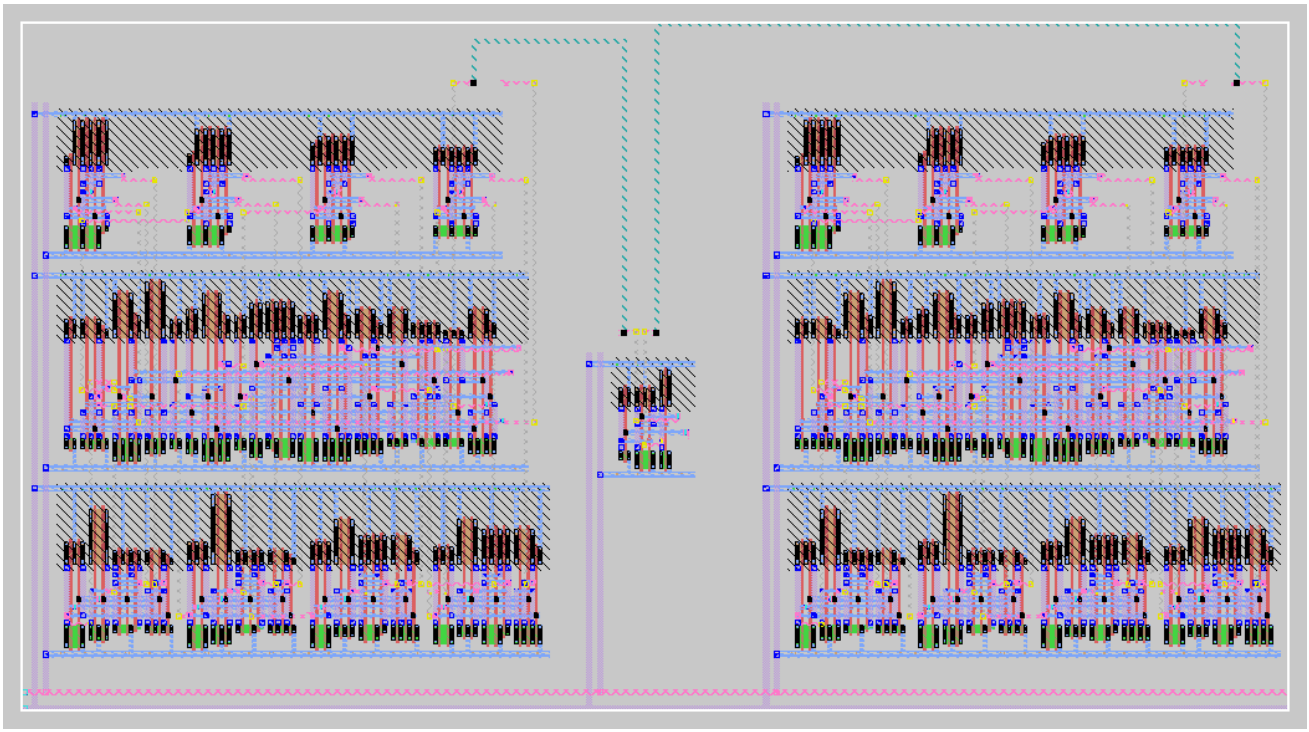


Fig. 47. SIMD 8-bit adder in MAGIC: CLA0, the gating cell and CLA1. The dashed lines at the top are metal 5.

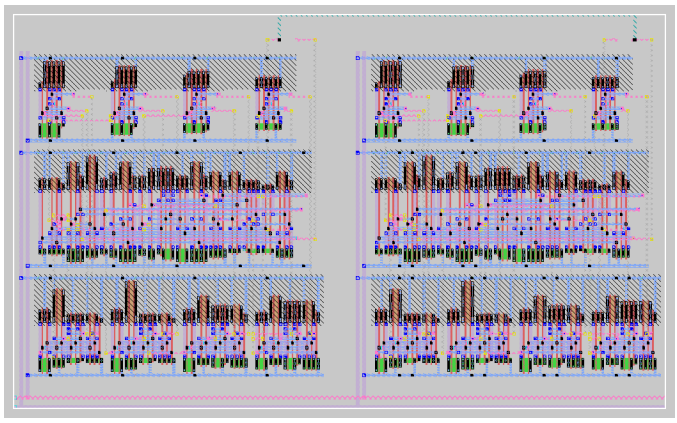


Fig. 49. Baseline 8-bit adder in MAGIC: c4lo runs on metal 5 straight from CLA0 to CLA1.

TABLE XVI
PART II LAYOUT RESULTS

Design	DRC	Devices (LVS)	Area (μm^2)
Baseline	0	388, match	7749.0
SIMD	0	396, match	8858.7

D. Worst-Case Delay in Both Modes

The search runs 13,312 single-input transitions per case: every transition of CLA0's nine inputs with CLA1 set to pass a carry through all four bits, and every transition of CLA1's eight inputs with CLA0's carry-out held at 0 and at 1.

TABLE XVII
WORST-CASE PROPAGATION DELAY

Case	Pre-layout (ps)	Post-layout (ps)
Baseline	855.22	1030.86
SIMD, mode 0	920.09	1109.77
SIMD, mode 1	516.31	570.22

In mode 0 the worst path is a3 fall to S5, through CLA0's carry tree and the gating cell into CLA1. In mode 1 the carry never crosses, so the worst path stays inside one 4-bit block.

E. Delay Penalty

Worst-case carry vector A = 0xFF, B = 0x00, $c_{in,0}$ toggling, delay from $c_{in,0}$ to S_7:

$$\text{Delay penalty} = \frac{t_{pd,SIMD} - t_{pd,baseline}}{t_{pd,baseline}} \times 100\%$$

TABLE XVIII
CIN TO S7 ON THE WORST-CASE VECTOR

View	Baseline (ps)	SIMD m0 (ps)	Penalty
Pre-layout	633.33	691.62	+58.28 ps, 9.20%
Post-layout	816.11	877.83	+61.72 ps, 7.56%

The gate alone takes 110.46 ps, but the penalty is only 61.72 ps: in the baseline, CLA0's carry-out drives CLA1's carry-in load directly, while in the SIMD design the gate isolates that load, so CLA0's carry-out switches sooner.

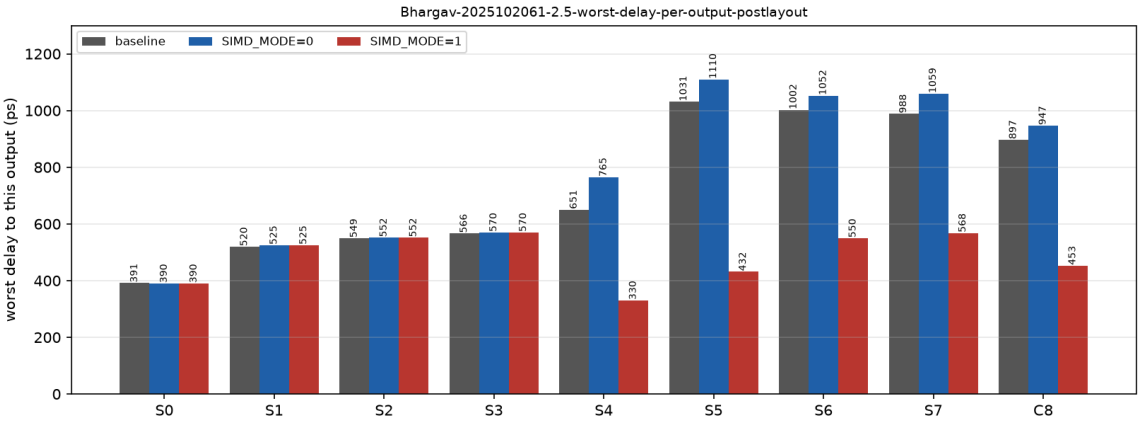


Fig. 50. Worst post-layout delay to each output.

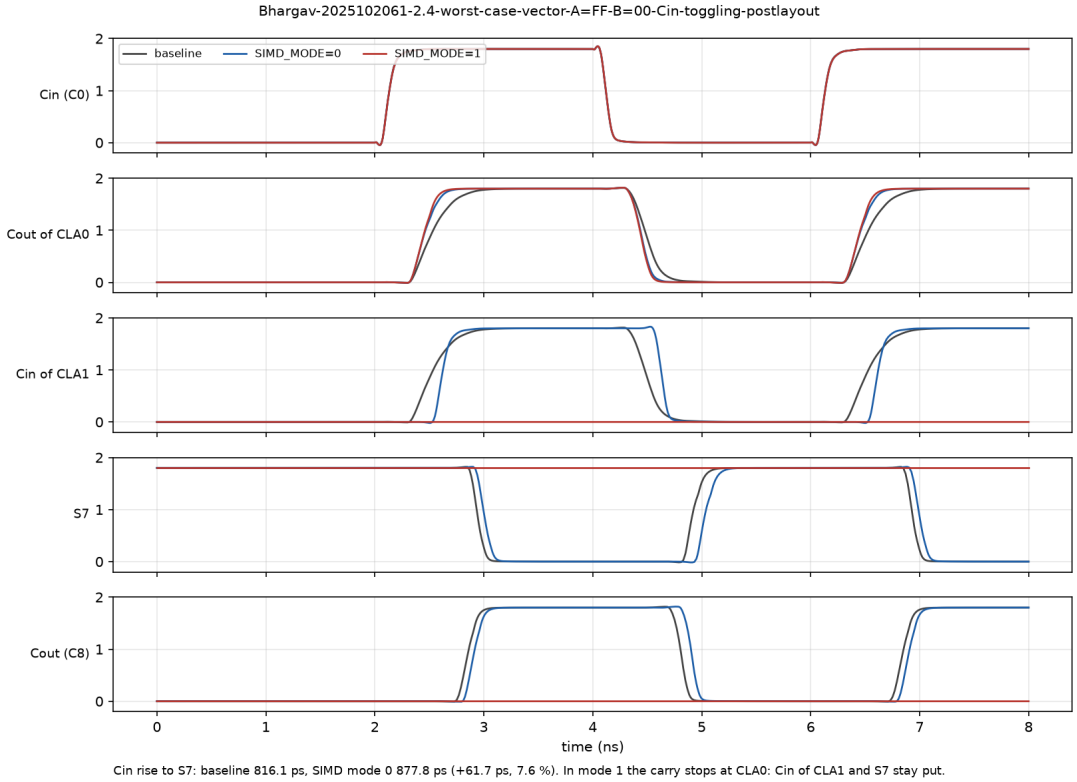


Fig. 51. Post-layout waveforms on the worst-case vector. In mode 1 CLA1's carry-in stays at 0.

F. Power Penalty

Course-formula f_{max}, with the Part I flip-flop's post-layout t_{su} and t_{hold}: baseline 0.897 GHz, SIMD mode 0 0.838 GHz, mode 1 1.529 GHz. At those clocks some random vectors that change several inputs at once did not settle, so the shortest clock at which all three cases settle on the same 200 random vectors was found: 1400 ps. Power is compared there, with identical inputs for all three.

$$\text{Power penalty} = \frac{P_{SIMD,0} - P_{baseline}}{P_{baseline}} \times 100\%$$

TABLE XIX
AVERAGE POWER AT 1400 PS, POST-LAYOUT

Case	10 cycles (μW)	200 vectors (μW)
Baseline	1695.2	1876.3
SIMD, mode 0	1720.7	1919.5
SIMD, mode 1	1592.4	1713.7

Power penalty in 8-bit mode: **1.50%** over 10 cycles, and 2.30% over 200 vectors.

G. Energy-Delay Product

$$EDP = P_{avg} \times t_{pd}^2$$

TABLE XX
ENERGY-DELAY PRODUCT, POST-LAYOUT

Case	P_{avg} (μW)	$t_{pd,max}$ (ps)	EDP (J s)
Baseline	1695.2	1030.86	1.801e-21
SIMD, mode 0	1720.7	1109.77	2.119e-21
SIMD, mode 1	1592.4	570.22	5.178e-22

TABLE XXI
COST OF RECONFIGURABILITY, SIMD MODE 0 AGAINST THE BASELINE, POST-LAYOUT

Measure	Penalty
Cin to S7 delay	7.56%
$t_{pd,max}$	7.65%
Power, 10 cycles	1.50%
EDP	17.6%
Transistors	+8
Area	14.3%

Delay counts twice in EDP, so the delay penalty dominates the small power penalty. In dual mode each add is only 4 bits long, so its EDP is far lower.

XII. VERILOG HDL

To be completed in the next draft, with Part III: behavioural Verilog of the 4-bit CLA and the SIMD adder, test vectors for every mode, and annotated simulation waveforms.

XIII. FPGA IMPLEMENTATION

To be completed in the next draft, with Part III: synthesis and implementation on the Boolean Board, test scenarios, and oscilloscope captures.

ACKNOWLEDGMENT

I thank Prof. Abhishek Srivastava for the course and this project, and the teaching assistants for their guidance.

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[7] M. Morris Mano, *Digital Logic and Computer Design*.

APPENDIX: NETLISTS

Generated netlists; widths and lengths in metres. All files are in the project folders `part1/spice/` and `part2/schematics/`.

4-bit CLA, top level

```
.subckt cla_final a0 a1 a2 a3 b0 b1 b2 b3 c0 s0 s1 s2 s3 c4
+ vdd gnd
Xnand0 a0 b0 g0b vdd gnd nand2_x1p25
Xnor0 a0 b0 t0b vdd gnd nor2_x1p56
Xaoi0 a0 b0 t0b p0 vdd gnd aoi21_x0p41
Xoai0 a0 b0 g0b p0b vdd gnd oai21_x0p41
Xnand1 a1 b1 g1b vdd gnd nand2_x1p25
Xnor1 a1 b1 t1b vdd gnd nor2_x1p95
Xaoi1 a1 b1 t1b p1 vdd gnd aoi21_x0p41
Xoai1 a1 b1 g1b p1b vdd gnd oai21_x0p41
Xnand2 a2 b2 g2b vdd gnd nand2_x1p25
Xnor2 a2 b2 t2b vdd gnd nor2_x1p25
Xaoi2 a2 b2 t2b p2 vdd gnd aoi21_x0p8
Xoai2 a2 b2 g2b p2b vdd gnd oai21_x0p8
Xnand3 a3 b3 g3b vdd gnd nand2_x1
Xnor3 a3 b3 t3b vdd gnd nor2_x1p25
Xaoi3 a3 b3 t3b p3 vdd gnd aoi21_x1
Xoai3 a3 b3 g3b p3b vdd gnd oai21_x1
Xinvc0 c0 c0b vdd gnd inv_x1
Xg10 t1b g0b g1b g1_0 vdd gnd oai21_fn_x1p25
Xt10 t1b t0b t1_0 vdd gnd nor2_x1p56
Xg32 t3b g2b g3b g3_2 vdd gnd oai21_x0p8
Xt32 t3b t2b t3_2 vdd gnd nor2_x0p8
Xig2 g2b g2 vdd gnd inv_x1p25
Xit2 t2b t2 vdd gnd inv_x1p95
Xg30 t3_2 g1_0 g3_2 g3_0b vdd gnd aoi21_fnp_x0p41
Xt30 t3_2 t1_0 t3_0b vdd gnd nand2_x0p41
Xg20 t2 g1_0 g2 g2_0b vdd gnd aoi21_fnp_x1
Xt20 t2 t1_0 t2_0b vdd gnd nand2_x1p25
Xig10 g1_0 g1_0b vdd gnd inv_x1
Xit10 t1_0 t1_0b vdd gnd inv_x1p56
Xc1 t0b c0b g0b c1 vdd gnd oai21_x0p51
Xc2 t1_0b c0b g1_0b c2 vdd gnd oai21_fn_x1p25
Xc3 t2_0b c0b g2_0b c3 vdd gnd oai21_fn_x1p25
Xc4g t3_0b c0b g3_0b c4 vdd gnd oai21_fnp_x0p8
Xx0 p0 p0b c0 s0 vdd gnd xor_aoi22_x1p25_0p51
Xx1 p1 p1b c1 s1 vdd gnd xor_aoi22_x1_0p64
Xx2 p2 p2b c2 s2 vdd gnd xor_aoi22_x0p8_1p25
Xx3 p3 p3b c3 s3 vdd gnd xor_aoi22_x0p51_1
.ends cla_final
```

Carry-gating cell

```
.subckt carry_gate_n1_i2 cout_lo mode cin_hi vdd gnd
Xinvm mode modeb vdd gnd inv_x1
Xnandg cout_lo modeb gn vdd gnd nand2_x1
Xinvg gn cin_hi vdd gnd inv_x2
.ends carry_gate_n1_i2
```

SIMD 8-bit adder, top level

```
.subckt simd8 a0 a1 a2 a3 a4 a5 a6 a7 b0 b1 b2 b3 b4 b5 b6
+ b7 c0 mode s0 s1 s2 s3 s4 s5 s6 s7 c8 c4lo vdd gnd
Xcla0 a0 a1 a2 a3 b0 b1 b2 b3 c0 s0 s1 s2 s3 c4lo vdd gnd
+ cla4_final
Xgate c4lo mode cinh vdd gnd carry_gate_n1_i2
Xcla1 a4 a5 a6 a7 b4 b5 b6 b7 cinh s4 s5 s6 s7 c8 vdd gnd
+ cla4_final
.ends simd8
```